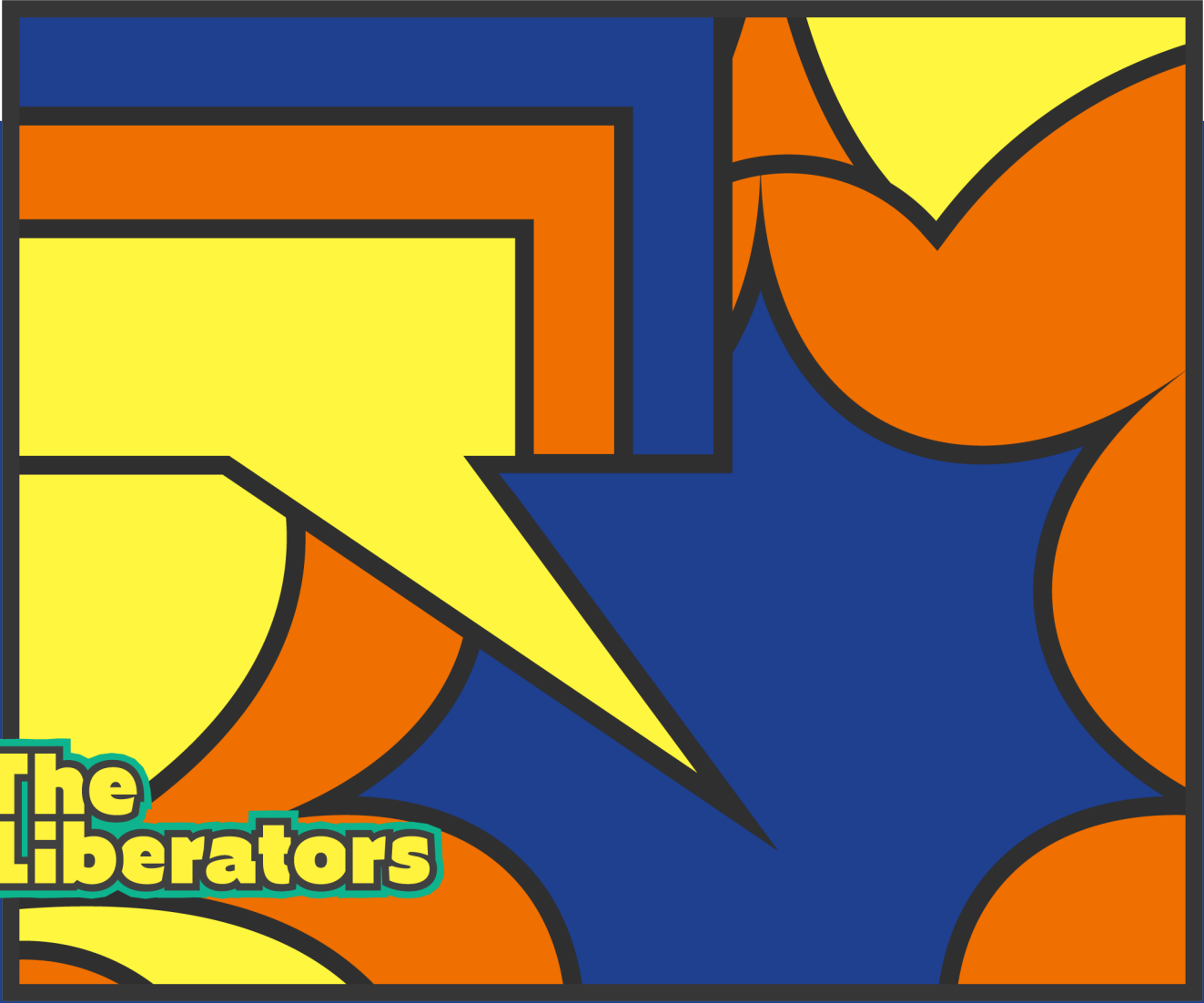


Map Dependencies To Find Bottlenecks

This simple experiment drives continuous improvement by keeping a close eye on where your external dependencies lead to bottlenecks

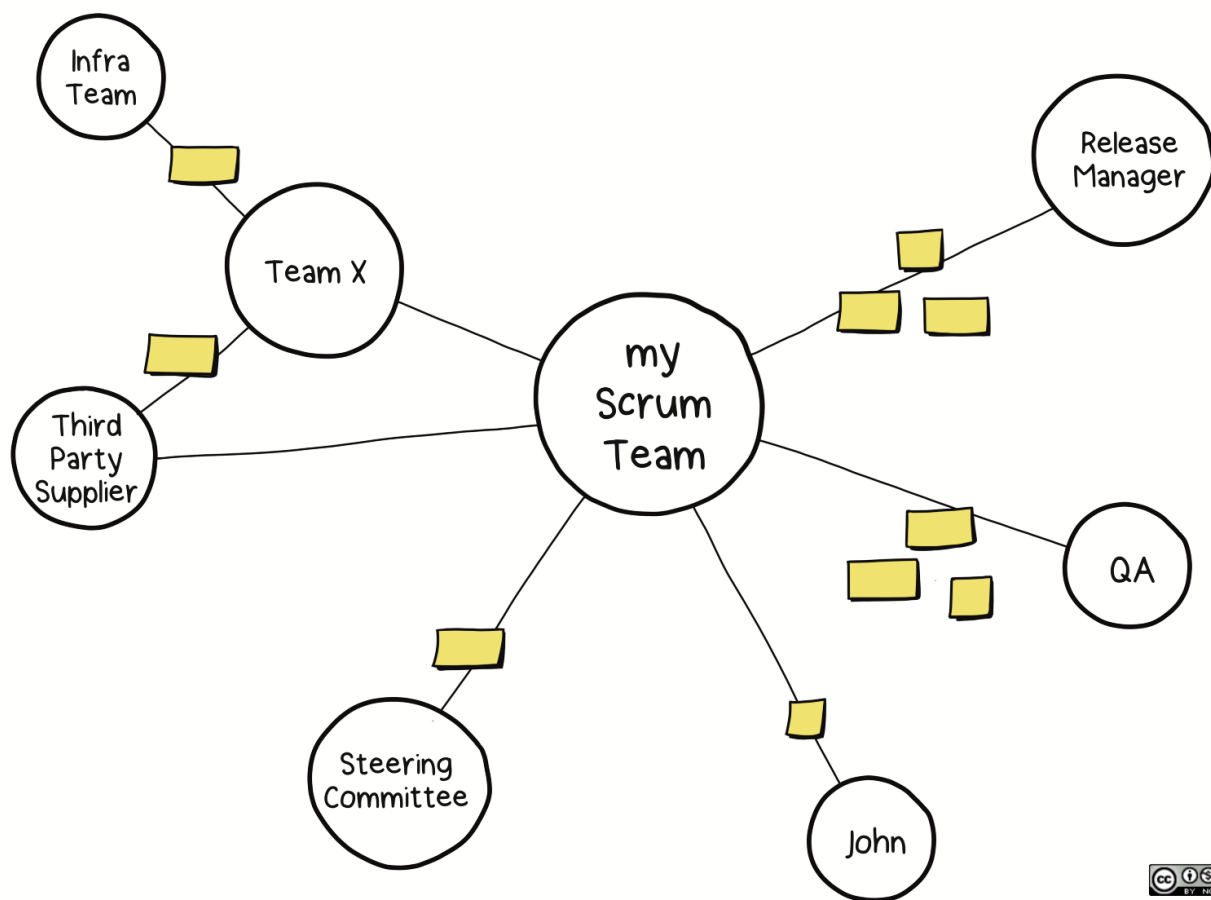
An abstract graphic featuring a dark blue background with a large, irregular shape composed of orange and yellow segments. The shape has a jagged, organic form, resembling a stylized letter 'L' or a map of a region. The orange segments are separated by yellow segments, creating a complex, interlocking pattern. The entire graphic is framed by a thick black border.

**The
Liberators**

Purpose Of This Experiment

It is hard for a Scrum team to ship fast when it has to wait on other departments, teams, or suppliers to do something they depend on. For example, when deployments are performed by an external team that is swamped with other work. Or when another department has to perform specialized testing before approval is given. Whatever their dependencies, they are generally outside of the control of a team. That makes the delays unpredictable. Even when a Scrum team considers something "Done", weeks or months may pass before their work actually reaches stakeholders. This greatly impedes a team's ability to work empirically and reduce the risk associated with complex work.

This experiment is about creating transparency around dependencies and the effect they have on your team's ability to ship fast. This experiment was inspired by the Dependency Spiders in Jimmy Janlen's ["96 Visualization Examples"](#) (which contains many other useful visualizations).



Created by Thea Schukken for the Zombie Scrum Survival Guide
by Christiaan Verwijs, Johannes Scharlau & Barry Overeem
zombiescrum.org

Difficulty / Effort



Required Skill



Mapping dependencies and calculating wait time are easy.

Impact On Survival



Removing or reducing dependencies is a great way to ship faster.

How To Prepare



Group Size

This string of Liberating Structures is feasible from 4 participants and onward. You can also run it with many teams simultaneously.

Intended Audience

The invitations and steps for this particular string are aimed at individual Scrum teams. The Scrum Team Survey also allows you to invite stakeholders to give their perspectives. If the team is up for it, you can also invite stakeholders to participate in this workshop – but only when they've filled out the survey as well.

Virtual Delivery

Liberating Structures work well virtually. Here too, they allow you to unleash and involve everyone. What is exceptionally important, however, is that you have the right tools. Because Liberating Structures are fundamentally about breaking down large-group interactions into smaller groups, you need tools that allow you to create subchannels or breakout rooms easily. [Zoom](#) is ideal for this, but similar features are available in tools like [Teams](#) and [GotoMeeting](#). We also recommend a virtual whiteboard where you can collect shared insights, like [Mural](#), [Miro](#), or [Google Slides](#). Even [Slack](#) can work.

Material

- Go to <https://survey.zombiescrum.org> and fill out the extensive free survey for your Scrum Team. After you have completed the first survey invite your team to join your 'sample' as instructed by the site. You can also invite stakeholders.
- Print the team-level report and hand them out, put prints on the walls or simply put up the profile on a screen. You can hide your personal responses in the profile with the toggle so that you don't share your own responses accidentally.



Steps



To implement this experiment, do the following:

1. Draw your Scrum team in the middle of a big piece of paper. Together, create a list of the teams, people, and departments you frequently need something from in order to create a Done Increment or release it. Whose approval do you need? Who needs to perform an activity in order for your team to continue? Draw the sources that you depend on around your team, like the legs of a spider.
2. Whenever your team needs something from someone outside the team, capture the request and the date it was issued on a sticky note and put it next to the source on the canvas. When the request is fulfilled, write the number of days you had to wait on the sticky. At the end of the Sprint, calculate the average wait time in days for all the requests that have been fulfilled, and move them to an archive.
3. Use the Dependency Spider and the average wait time as input for your Sprint Reviews and Sprint Retrospectives. What actions can you take to reduce the impact of dependencies on your ability to ship? How can you include and collaborate with them to remove or reduce dependencies? How can you leverage support from your Product Owner and stakeholders to change the environment of your Scrum team so that you can ship value to them faster?

Our Findings



- Dependency Spiders are a wonderful tool to visualize what is preventing the system from becoming more effective. You may find that the people you depend on also have dependencies of their own. Although it can be simpler to limit your analysis to the dependencies of your team, you will get a more complete picture of the overall dependencies by inviting other teams, people, and departments to participate.



Experience From The Field



"We used Dependency Spiders to understand a huge dependency that was impeding our ability to ship a large infrastructure product. Releasing new Increments was outsourced to a service provider. Because they used a different schedule and worked for many other customers, our requests would end up somewhere down a lengthy backlog. Not only did we have to wait an unpredictable number of weeks for something to happen, when they finally got around to it they'd bombard us with questions and issues that disrupted our focus from the current Sprint.

After creating awareness around the wait time, and the impact it was having on our focus and ability to get feedback from stakeholders, the Product Owner worked with management and the service provider to assign a part-time release engineer to each of the five Scrum Teams. Although there was no requirement to be physically present, we noticed that most of the release engineers made an effort to be with the teams in person."

More Resources

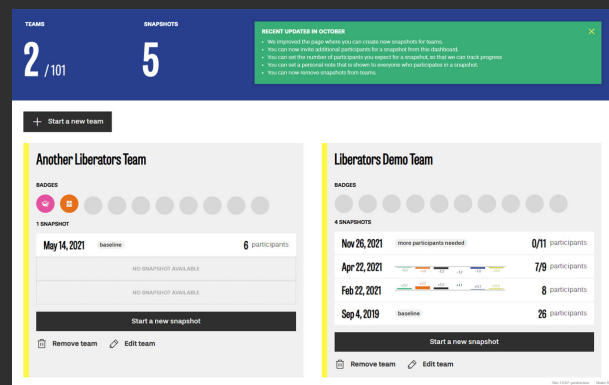
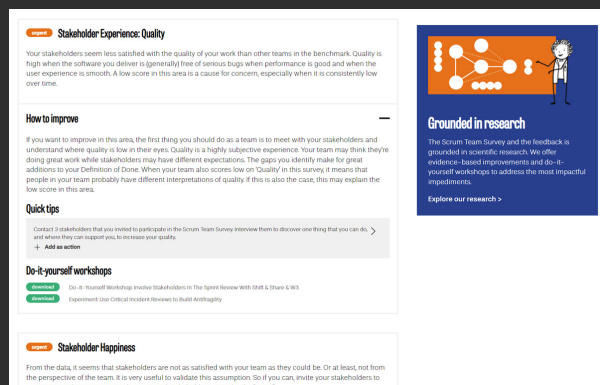
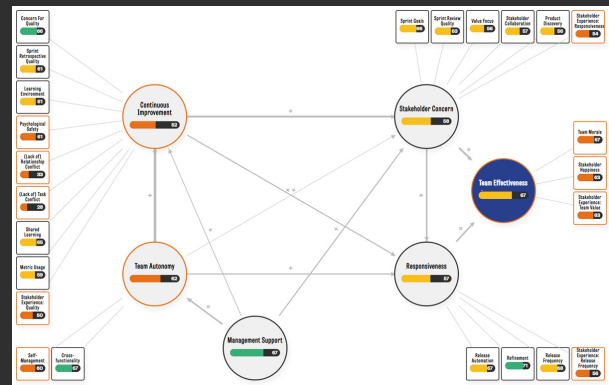
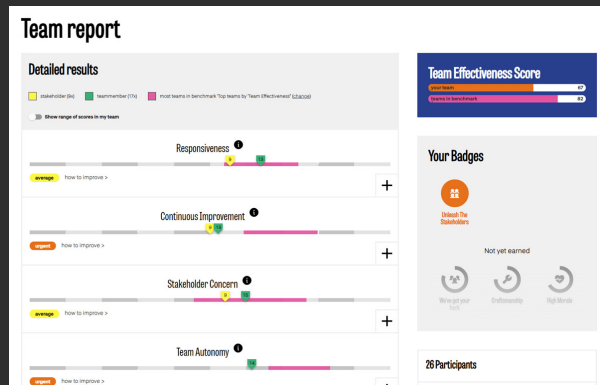


- Whitepaper: [The 8 Stances of a Scrum Master](#) illustrates well why Scrum Masters are responsible for removing certain impediments.
- Research: [How Do Scrum Masters Deal With Problems Outside Of Their Team's Control?](#). Together with [Age of Product](#), we asked Scrum Masters how they overcame impediments (including dependencies).
- Podcast: [Scrum Mythbusters: The Scrum Master Must Resolve Every Problem](#)



Make Your Scrum Team More Effective

Want to make your Scrum team more effective? We created the Scrum Team Survey to help. Invite members and stakeholders to a detailed questionnaire and diagnose the results together. You receive tons of actionable feedback based on your results, including dozens of quick tips and useful do-it-yourself workshops. You can use the Scrum Team Survey for free with your team. No account or e-mail address is required. You can also subscribe for more advanced features.



	Essential	Liberator
Invite team members & stakeholders	unlimited	unlimited
Quick Tips for your team per topic	3	unlimited
Recommendations to your team	5	unlimited
Compare your team against different benchmarks		✓
Analyze the range of scores in your team		✓
Receive our personal guidance on how to get started		✓
See where your team improves over time with Team Trends		✓
Support our academic research to help more teams		✓
Pricing per team per month	€ 0	€ 10

<https://scrumteamsurvey.org>

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